Manipulating and Visualizing Qubit States

Lab Handout

Session Overview

Duration: 1h15m

Session Objectives

- Visualize qubit states using Bloch sphere (poles=computational basis, equator=superposition).
- Apply single-qubit gates (H, X, Z, RX/RY/RZ) and track Bloch sphere rotations.
- Interpret q-sphere visualizations showing amplitude magnitudes + complex phases.
- Construct Bell states and understand multi-qubit statevectors.
- Predict gate effects before running circuits (training geometric intuition).

Timed Agenda:

- 0-10min: Statevectors & Bloch basics
- 10-30min: Single-qubit gates (H, X, Z, rotations)
- 30-50min: Multi-qubit states & entanglement
- 50-70min: Advanced visualizations (q-sphere)
- 70-75min: Exercises

1. Setup

```python
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector, state_fidelity
from qiskit_aer import AerSimulator
from qiskit.visualization import plot_bloch_multivector, plot_state_qsphere,
    plot_histogram
import matplotlib.pyplot as plt
import numpy as np

sim = AerSimulator()
print('Qiskit visualization tools ready!')
```

2. Qubit States & Bloch Sphere (10min)

A single qubit state is written in the computational basis as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

Key Question: How do we map (α, β) to a 3D point $\vec{r} = (x, y, z)$ on the unit sphere?

The Bloch Vector Formula

The explicit mapping is:

$$\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2\,\mathrm{Re}(\alpha^*\beta) \\ 2\,\mathrm{Im}(\alpha^*\beta) \\ |\alpha|^2 - |\beta|^2 \end{pmatrix}$$

Standard States

The complete set of cardinal states in the computational basis:

$$|0\rangle = |0\rangle \qquad (+\text{Z eigenstate})$$
$$|1\rangle = |1\rangle \qquad (\text{-Z eigenstate})$$
$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad (+\text{X eigenstate})$$
$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \qquad (\text{-X eigenstate})$$
$$|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}} \qquad (+\text{Y eigenstate})$$
$$|-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}} \qquad (\text{-Y eigenstate})$$

Final Bloch sphere table (all 6 directions):

State	α	β	Bloch Vector	Position
$\|0\rangle$	1	0	$(0, 0, +1)$	+Z (North pole)
$\|1\rangle$	0	1	$(0, 0, -1)$	-Z (South pole)
$\|+\rangle$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	$(+1, 0, 0)$	+X equator
$\|-\rangle$	$\frac{1}{\sqrt{2}}$	$\frac{-1}{\sqrt{2}}$	$(-1, 0, 0)$	-X equator
$\|+i\rangle$	$\frac{1}{\sqrt{2}}$	$\frac{i}{\sqrt{2}}$	$(0, +1, 0)$	+Y equator
$\|-i\rangle$	$\frac{1}{\sqrt{2}}$	$\frac{-i}{\sqrt{2}}$	$(0, -1, 0)$	-Y equator

Physical Picture

- **Latitude** (z): Set by population imbalance $|\alpha|^2 : |\beta|^2$
- **Longitude** (ϕ): Set by relative phase $\arg(\beta/\alpha)$
- **Surface**: All pure states lie on the **unit sphere** ($|\vec{r}| = 1$)

```
# |0>: North pole
sv0 = Statevector.from_label('0')
print(f"|0>: {sv0.data}")
plot_bloch_multivector(sv0)

# |+>: Equator (+X axis)
qc_h = QuantumCircuit(1); qc_h.h(0)
sv_plus = sv0.evolve(qc_h)
print(f"|+>: {sv_plus.data}")
plot_bloch_multivector(sv_plus)
```

3. Single-Qubit Gates (20min)

Gate Effects on Bloch Sphere

Gate	$\|0\rangle \rightarrow$	Bloch Before	Bloch After
X	$\|1\rangle$	North pole	South pole
H	$\|+\rangle$	North pole	+X equator
Z	$e^{i\pi}\|0\rangle$	North pole	North pole (phase invisible)
RX($\pi/2$)	$\|-i\rangle$	North pole	-Y equator
RY($\pi/2$)	$\|+\rangle$	North pole	+X equator

```
# X gate: North -> South pole
qc_x = QuantumCircuit(1); qc_x.x(0)
sv_x = sv0.evolve(qc_x)
plot_bloch_multivector(sv0, title="X Gate Before"); plot_bloch_multivector(sv_x
    , title="X After")

# H gate: North -> +X equator
sv_h = sv0.evolve(QuantumCircuit(1).h(0))
plot_bloch_multivector(sv0, title="H Gate Before"); plot_bloch_multivector(sv_h
    , title="H After")

# Z gate: Phase flip (invisible on Bloch)
sv_z = sv0.evolve(QuantumCircuit(1).z(0))
plot_bloch_multivector(sv0, title="Z Before"); plot_bloch_multivector(sv_z,
    title="Z After")
```

RX($\pi/2$) Example The RX($\pi/2$) gate matrix is:

$$RX(\pi/2) = \begin{pmatrix} \cos(\pi/4) & -i\sin(\pi/4) \\ -i\sin(\pi/4) & \cos(\pi/4) \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \tag{1}$$

Apply to initial state $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$RX(\pi/2)|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \cdot 1 + (-i) \cdot 0 \\ -i \cdot 1 + 1 \cdot 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \tag{2}$$

Thus: $\alpha = \frac{1}{\sqrt{2}}$, $\beta = \frac{-i}{\sqrt{2}}$

Bloch vector computation $\vec{r} = (x, y, z)$ where:

$$x = 2\,\mathrm{Re}(\alpha^*\beta) = 2\,\mathrm{Re}\left(\frac{1}{\sqrt{2}} \cdot \left(-\frac{i}{\sqrt{2}}\right)\right) = 2\,\mathrm{Re}\left(-\frac{i}{2}\right) = 2 \cdot 0 = 0 \tag{3}$$

$$y = 2\,\mathrm{Im}(\alpha^*\beta) = 2\,\mathrm{Im}\left(\frac{1}{\sqrt{2}} \cdot \left(-\frac{i}{\sqrt{2}}\right)\right) = 2\,\mathrm{Im}\left(-\frac{i}{2}\right) = 2 \cdot \left(-\frac{1}{2}\right) = -1 \tag{4}$$

$$z = |\alpha|^2 - |\beta|^2 = \frac{1}{2} - \frac{1}{2} = 0 \tag{5}$$

Result: $\vec{r} = (0, -1, 0)$ = **-Y pole**!

$$\mathrm{RX}(\pi/2)\text{: } |0\rangle \xrightarrow{\text{matrix}} \frac{|0\rangle - i|1\rangle}{\sqrt{2}} \xrightarrow{\text{Bloch}} (0, -1, 0)$$

Live Demo: Interactive Rotations

```
# Predict before running! Edit \theta
theta = np.pi/4  # Try pi/6, pi/3
qc_custom = QuantumCircuit(1); qc_custom.rx(theta, 0)
plot_bloch_multivector(sv0.evolve(qc_custom), title=f'RX({theta:.2f})')
```

4. Q-Sphere: Complex Phases (10min)

Q-sphere shows amplitude $|\alpha|$ (size) + phase $arg(\alpha)$ (color). Essential for multi-qubit states.

```
# H|0> vs H|1> phase difference
qc_h0 = QuantumCircuit(1); qc_h0.h(0); sv_h0 = Statevector.from_instruction(
    qc_h0)
qc_h1 = QuantumCircuit(1); qc_h1.x(0); qc_h1.h(0); sv_h1 = Statevector.
    from_instruction(qc_h1)
plot_state_qsphere(sv_h0.data); plot_state_qsphere(sv_h1.data)
```

Key insight: $\mathrm{H}|1\rangle = |-\rangle$ appears opposite $\mathrm{H}|0\rangle = |+\rangle$ on the Bloch sphere because the Hadamard introduces a 180° relative phase.

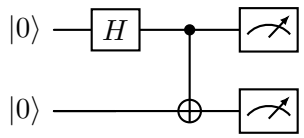
$$\mathrm{H}|0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle \quad (1, 0, 0)$$

$$\mathrm{H}|1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle \quad (-1, 0, 0)$$

Phase effect: The $\beta = -1/\sqrt{2}$ coefficient gives $\alpha^*\beta = -1/2 \rightarrow x = 2\,\mathrm{Re}(-1/2) = -1$, flipping from +X to -X pole!

5. Multi-Qubit States & Entanglement (20min)

Bell state: $\Phi^+ = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ (perfect correlation).




```
# Bell pair statevector
qc_bell = QuantumCircuit(2); qc_bell.h(0); qc_bell.cx(0,1)
sv_bell = Statevector.from_label('00').evolve(qc_bell)
print(f"Bell: {sv_bell}")
plot_state_qsphere(sv_bell.data)  # Two equal spheres (00,11), phase=0
```

6. Advanced Visualizations (10min)

Compare multiple states:

```
qc_states = [QuantumCircuit(1).h(0),  # |+>
             QuantumCircuit(1).x(0).h(0),  # |->
             QuantumCircuit(1).ry(np.pi/2,0)]  # RY|0>
plot_bloch_multivector(qc_states)
```

7. Homework Exercises (5min)

Ex1 Create $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$ (should be -X equator).

Ex2 Predict q-sphere for HZH$|0\rangle$.

```
# Ex1 Template - FILL IN
qc_ex1 = QuantumCircuit(1)
# ...
sv_ex1 = Statevector.from_label('0').evolve(qc_ex1)
plot_bloch_multivector(sv_ex1); print(sv_ex1)
```

Quantum Gates Summary

Gate	Matrix	$\|0\rangle \to$	Bloch Sphere Effect
H	$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$	$\|+\rangle = \frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$	North pole $\to$ +X equator
X	$\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$	$\|1\rangle$	North $\to$ South pole
Z	$\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$	$\|0\rangle$ (phase)	+Y $\to$ -Y equator
RX($\pi/2$)	$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & -i\\ -i & 1\end{pmatrix}$	$\|-i\rangle = \frac{\|0\rangle - i\|1\rangle}{\sqrt{2}}$	North $\to -$Y equator
RY($\pi/2$)	$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & -\sin(\pi/4)\\ \sin(\pi/4) & 1\end{pmatrix}$	$\|+\rangle$ (approx)	North $\to$ +X equator
RZ($\pi/2$)	$\begin{pmatrix}1 & 0\\ 0 & -i\end{pmatrix}$	$\frac{\|0\rangle - i\|1\rangle}{\sqrt{2}}$	Z-phase rotation

Key Takeaways

- **Bloch sphere:** Pure single-qubit states as 3D vectors (gates=rotations).
- **Q-sphere:** Multi-qubit amplitudes + phases (size=color).
- **Predict first:** Sketch Bloch position before running code.
- **Phase invisible:** Z-basis measurement misses Y/X phase info.
- **Entanglement geometry:** Bell state lives in 4D ($\mathbb{C}^4$) but q-sphere reveals structure.